



Areeb Asgar
Mechanical Engineering
Indian Institute of Technology Bombay

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B.Tech.
Gender: Male
DOB: 07/12/2002

Examination	University	Institute	Year	CPI / %
Graduation	IIT Bombay	IIT Bombay	2025	
Intermediate	CBSE	Sunbeam English School, Bhagwanpur, Varanasi	2021	98.40%
Matriculation	CBSE	Sunbeam English School, Bhagwanpur, Varanasi	2019	96.60%

Pursuing a **Minor Degree** in the department of **Computer Science and Engineering** at IIT Bombay

SCHOLASTIC ACHIEVEMENTS

- Secured **98.5** percentile in JEE Mains among more than **0.93 million** candidates across all over India [‘21]
- Ranked among top **2.7** percentile in JEE Advanced from over **0.14 million** students across all over India [‘21]

PROFESSIONAL EXPERIENCE

Software Development Engineer Intern [May ‘24 – July ‘24]

Amazon Development Centre

- Encrypted the sensitive carrier data stored in **DynamoDB** at the persistence layer with an Attribute Transformer
- Migrated the Tax Details service to the new workflow, upgrading over **7** dependencies to more stable versions
- Integrated utility for gradual percentage based rollout of feature to users for smooth migration and easy rollback
- Developed **automated tests** for the development stage deployments to reduce risk of bugs in production stages
- Added timeout configurations of **JLB relay** to handle high latency requests to service and reduce average latency
- Rectified the flow of requests to new workflow, instead of the legacy workflow, for all ship methods of **2** carriers

Application Development Intern [July ‘23 – Oct ‘23]

Upayahealth Pvt. Ltd.

Providing healthcare screening and triaging services to the rural population of India with little access to healthcare

- Employed Flutter and Django to create **healthcare triaging** mobile applications to provide early stage screening
- Incorporated **multi-language** support to cater to the diverse population of India, beginning with **2** languages
- Encrypted sensitive user data prior to storage in order to adhere to the **HIPAA** guidelines for healthcare data
- Integrated third party APIs for AI triaging tool to generate screening report for the users, sent to them via SMS

COMPETITIONS AND INTERNATIONAL EXPOSURE

Robo World Cup 2024 | Federation of International Robot-soccer Association [Jan ‘24 – Aug ‘24]

Unmesh Mashruwala Innovation Cell (UMIC), IIT Bombay

- Led a contingent of 5 members into the final round of the Autonomous Cars (Pro) challenge at **Sao Luis, Brazil**
- Supervised development of 1:10 scale autonomous car that can intelligently navigate in race and urban settings

Flipkart Grid 5.0 Robotics Challenge [July ‘23 – Jan ‘24]

Unmesh Mashruwala Innovation Cell (UMIC), IIT Bombay

- Emerged as the **National Champions** of Robotics Track among 21 finalist teams, selected from **25K** participants
- Led a team of 6 to construct a **5 DOF** robotic arm capable of autonomous singulation for 3 classes of packages

KEY PROJECTS

SeDriCa [Feb ‘22 – Present]

Unmesh Mashruwala Innovation Cell (UMIC), IIT Bombay

Working in team of 30+ students aiming to develop India's First Level 5 Self Driving Car, customised for Indian roads

- Exploring fast and robust solutions to the **Simultaneous Localisation and Mapping** problem using **LiDAR** sensors
- Employing state of the art algorithms like **ORB-SLAM** and **LOAM** to localise the vehicle and map its environment
- Configured **Extended Kalman Filter** Package to fuse **GPS, IMU** data to derive state vector for real time position
- Curated a swift variant of **Iterative Closest Point** algorithm to estimate real-time location using point cloud data

Course Project | Guide: Prof. S. Sudarshan

- Implemented caching at database layer for a remote **PostgreSQL** database to improve application performance
- Achieved faster lookups on select operations for applications using caching, implemented with **libpq** library in C

Buffer Overflow in Heap | Principles of Data and System Security

[Apr '23 – May '23]

Course Project | Guide: Prof. Virendra Singh

- Exploited overflow due to unchecked input to a block in heap to expose potential for malicious code execution
- Implemented **virtual table hijacking**, in C++, for objects inside heap, successfully diverting program control flow
- Devised code for **corrupting metadata** of an allocated block, which, in consequence, returns an arbitrary pointer

Techpoints Portal

[Oct '22 – Nov '22]

Institute Technical Council (ITC), IIT Bombay

- Developed a portal to introduce techpoints, a revolutionary system for prize distribution at all institute events
- Constructed over **10** APIs and **3** admin pages using Django to store, process and display tech points of students
- Enabled IITB **SSO login** to authenticate users and authorise the portal only to registered students of IIT Bombay

Secure KNN Computation on Cloud | Advanced Network Security and Cryptography

[Dec '23]

Course Project | Guide: Prof. Virendra Singh

- Implemented **homomorphic encryption** on remote data to facilitate secure **K-Nearest Neighbours** computation
- Achieved accuracy of **10 cm** on real-life point cloud data for algorithm proposed in Secure KNN research paper

(De)Noise | Seasons of Code

[May '22 – July '22]

Web and Coding Club (WnCC), IIT Bombay

- Built a model to filter out background noises from an audio clip to generate the unblemished voice of a person
- Implemented Speech Enhancement GAN (**SEGAN**) and incorporated **28** speakers in **40** different noise conditions
- Achieved Mean Absolute Error of **0.00606** after extensively training the SEGAN model for speech enhancement

LangFusion Mobile Application | Introduction to Machine Learning

[Apr '24]

Course Project | Guide: Prof. Amit Sethi

- Created a mobile application to enable global chat among users by translating messages to preferred language
- Trained model based on the Google Neural Machine Translation (**GNMT**) on TED talks dataset with **60** languages

POSITIONS OF RESPONSIBILITY**Head of Operations**

[Apr '24 – Present]

Unmesh Mashruwala Innovation Cell (UMIC), IIT Bombay

- Overseeing logistics and progress of the team, enabling research activities and participation in global challenges
- Orchestrated the **two month** long recruitment of **150** aspirants, comprising of interviews, trainings and projects
- Forging relations with academic and industry experts, as well as sponsors to ensure latest R&D facilities in lab

Subsystem Lead, Localization

[Apr '23 – Mar '24]

Unmesh Mashruwala Innovation Cell (UMIC), IIT Bombay

- Managed one of the five subsystems of the team, tasked with providing real time location estimate of the car
- Responsible for preparing timeline for efficient operation of subsystem, allocating tasks and bringing new ideas
- Foremost support on budgeting proposals, recruitments, trainings and sponsorship for proper team execution

Web Convener

[May '22 – Mar '23]

Institute Technical Council (ITC), IIT Bombay

- Part of **6** membered team responsible for creating and maintaining websites of the Institute Technical Council
- Developed **3** portals from the ground up, to completion and further initiated **2** portals over the span of a year

TECHNICAL SKILLS

- **Programming** C++ | Python | Java | Kotlin | Dart | SQL | Linux | HTML | CSS | JavaScript | Git | GitHub
- **Development** Django | MySQL | Bootstrap | Angular | Flutter | Google Firebase | Android Studio
- **Machine Learning** Numpy | Pandas | Matplotlib | Seaborn | Scikit-Learn | TensorFlow
- **Robotics** Robot Operating System (ROS) | Gazebo | Rviz | URDF

EXTRACURRICULAR ACTIVITIES

- Constructed a **wifi controlled bot** from scratch using **ESP-32** and **L293D** as a part of the XLR8 competition by ERC
- Completed a year-long **military training** program for freshmen under National Cadet Corps (**NCC**), IIT Bombay